

Design of Air Systems for Ice Control

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Background

Air systems operate by delivering compressed air to a submerged orifice which is discharged into the water column generating a bubble stream. The air bubbles rise due to buoyancy and impart a momentum flux on the water column. The rising water then entrains surrounding water and a plume of vertical flow grows laterally as it rises. The plume which consists of bubbles at its core surrounded by entrained water rises and eventually encounters the free water surface. There the vertical flow of water must transition to horizontal flow due to continuity and thus a horizontal current is formed at the surface.

Inducing both vertical and horizontal velocities in water body can be useful for controlling ice in navigation settings. The most common applications of air systems in ice control settings are to transport heat from the bottom of the water column to suppress ice formation and melt surface ice (vertical current) or to move or arrest the movement of ice on the surface (horizontal current).

Scope of an Air System Conceptual Design

The conceptual design of an air system for ice control consists of the following:

- 1) Dimensions of diffuser/manifold and supply pipes (lengths, diameters);
- 2) Spacing and diameter of the orifices;
- 3) System airflow rates with corresponding pressures

Because there are varying purposes for using air systems for ice control there is not a one-size-fits-all approach to designing system components for optimal performance in all cases. High-flow air screens or screens used to control the movement of ice should be designed to produce uniform surface currents at a velocity sufficient to hold back or deflect ice from driving forces. High-flow recess and point bubblers used to flush ice from gate recesses also produce surface currents. Still, they tend to operate at high intensity for short durations, in which case air demands are an important design consideration. Low-flow air bubblers used to move warmer water from the bottom of the water column are designed to generate sufficient vertical flow to create a gentle circulation but not so rigorous to exhaust the thermal reserve. Design considerations for these specific applications are addressed individually in later sections.

Velocity Fields in Bubble Induced Flow

The behavior of bubble-induced flow has seen much attention in the literature. Kobus (1968) presents an analytical solution for the velocity field assuming a Gaussian distribution with variation along a radial axis extending from the orifice. The plume spread is assumed to increase

linearly with increasing vertical distance from the orifice. Experimental observations suggest that this assumed distribution is accurate. Associated equations describing velocity and flow within the plume are relatively simple for design purposes (Ashton 1974, 1979).

Kobus divided the solutions for the velocity fields of bubble-induced plumes into point and line bubblers geometries.

Point Bubblers

In the ideal case of a single orifice discharging air in stagnant water, an axisymmetric plume is formed, which spreads linearly as it rises. In this case, the vertical and horizontal surface velocities are uniform in all directions for a given radial distance from the orifice. The distribution of vertical water velocity for a point bubbler from Kobus has the form

$$\frac{U(x, r)}{U_c(x)} = e^{-r^2/2c^2(x+x_0)^2} \quad (1)$$

where $U(x, r)$ = the vertical velocity in the plume as a function of x = vertical distance from the orifice and r = the radial distance from the orifice; U_c = the centerline velocity above the orifice; x_0 = distance from the analytical origin, which is a correction term accounting for the air jet's initial momentum and the complex flow behavior before the region of induced flow. Kobus (1968) determined x_0 was approximately 0.8 meters by experiment. The remaining term is c = the rate of lateral spread of the rising plume

$$c = C_c Q_o^{0.15} \quad (2)$$

where C_c = a coefficient measured experimentally ($0.152 \text{ m}^{-0.45} \text{ s}^{0.15}$). Q_o is the orifice airflow rate in cubic meters per second at standard temperature and pressure (SCMS). While definitions of “standard” temperature and pressure vary, in ice control air systems the convention has traditionally referenced a temperature of 0°C and atmospheric pressure (101.325 kPa), which is followed here. This is different than a common industry definition for standard cubic feet (or meters) per minute, SCFM (SCMM) which references 68°F.

The equation for the centerline velocity above the orifice for a point bubbler is given by

$$U_c(x) = \frac{1}{c(x+x_0)} \sqrt{\frac{-P_{atm} Q_o}{\pi \rho_w \bar{u}_b} \ln(1 - x/h^*)} \quad (3)$$

where P_{atm} = the atmospheric pressure; ρ_w = the density of water; $\bar{u}_b$ is the mean rising speed of the bubbles and h^* is the total head at the orifice located at a depth of h , including atmospheric pressure

$$h^* = h + \frac{P_{atm}}{\gamma_w} \quad (4)$$

where γ_w = the specific weight of water. At sea level $h^* = h + 10.3$ meters.

The mean rising speed of bubbles was measured by Kobus and the following empirical relationship is defined by Ashton (1979)

$$\bar{u}_b = C_b Q_o^{0.15} \quad (5)$$

where C_b = a coefficient measured experimentally ($1.83 \text{ m}^{0.55} \text{ s}^{-0.85}$).

The radial distance from the orifice, $b(x)$, corresponds the location of to the first standard deviation σ of the gaussian velocity distribution

$$b(x) = (x + x_0) C_c Q_o^{0.15} \quad (6)$$

which is linear with respect to depth and weakly dependent on the airflow rate.

Equations (1) and (3) can be combined to describe the magnitude and spatial distribution of the axisymmetric vertical velocity field as a function of the vertical distance above the orifice, x and the radial distance r from the orifice (Figure 1).

Line Bubblers

When orifices are placed close together in a line or row, such as on a manifold, the spread of individual plume overlap along the manifold axis. The horizontal surface velocities from adjacent plumes oppose each other along the manifold axis, therefore the resulting horizontal surface velocities are predominately perpendicular to the manifold axis.

The analytical solution for the line bubbler from Kobus (1968) is a limiting case that assumes that the orifices are infinitely close together. In other words, the analysis uses the limiting case of a continuous constant-width slot along the length of the manifold. This simplification yields an identical solution along the length of the manifold, with the only variation lies in the 2-dimensional plane perpendicular to the manifold. If the orifices are spaced too far apart or if the solution is applied in shallow water where the plumes may have limited spread, the assumption of uniform airflow per length of the manifold may not be appropriate. In these cases, the orifices may behave more like individual point bubblers with zones of weak vertical and horizontal surface velocity between them, which the analytical solution will not capture.

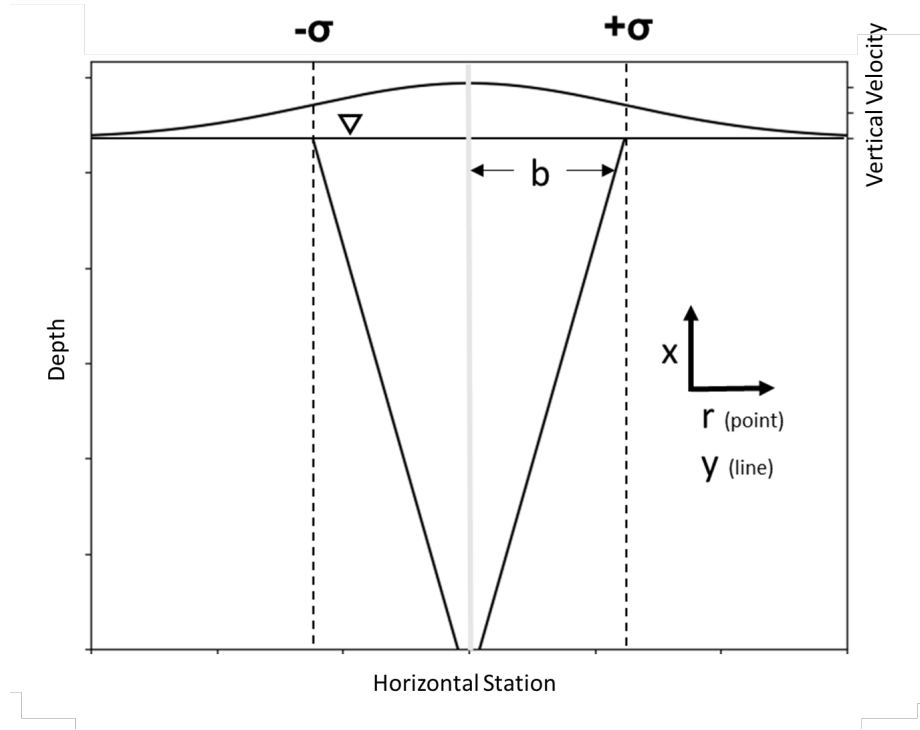


Figure 1: Definition drawing of an axis-symmetric and 2-dimensional bubbler plume, modified from Ashton (1979).

Vertical Water Velocity

The velocity field equations for line bubblers are similar to the point bubbler case, but there are some important differences. The solution is 2-dimensional in the x-y plane (vertical and perpendicular to the manifold), where airflow is expressed in terms of discharge per unit length. The experimentally determined coefficients for plume spread and average bubble velocity are slightly different than those for the point bubbler case.

$$\frac{U(x, y)}{U_c(x)} = e^{-y^2/2c^2(x+x_0)} \quad (7)$$

where $U(x, y)$ = the vertical velocity in the plume as a function of x = vertical distance from the orifice and y = the horizontal distance from the orifice (perpendicular to the manifold axis); U_c = the centerline velocity above the orifice; x_0 = distance from the analytical origin (0.8 meters from Kobus, 1968); c = the rate of lateral spread of the rising plume

$$c = C_c Q_a^{0.15} \quad (8)$$

Where C_c = a coefficient measured experimentally ($0.182 \text{ m}^{-0.3} \text{ s}^{0.15}$) and Q_a is the manifold airflow rate in cubic meters per second per meter at standard temperature and pressure (SCMS/m). $Q_a = Q_o/s$ where s is the orifice spacing and Q_o is the individual orifice airflow

rate (SCMS). The total airflow rate (SCMS) in the manifold can be defined as $Q_v = n \cdot s \cdot Q_a$ where n is the number of orifices.

The equation for the centerline velocity above the manifold for a line bubbler is given by

$$U_c(x) = \sqrt{\frac{-P_{atm} Q_a}{\sqrt{\pi} \rho_w \bar{u}_b c(x + x_0)} \ln(1 - x/h^*)} \quad (9)$$

The mean rising speed of bubbles was measured by Kobus (1968) and the following empirical relationship is defined by Ashton (1974)

$$\bar{u}_b = C_b Q_a^{0.15} \quad (10)$$

where C_b = a coefficient measured experimentally ($2.14 \text{ m}^{0.7} \text{ s}^{-0.85}$).

The perpendicular distance from the manifold, $b(x)$, corresponds the location of to the first standard deviation σ of the gaussian velocity distribution

$$b(x) = (x + x_0) C_c Q_a^{0.15} \quad (11)$$

which is linear with respect to depth and weakly dependent on the airflow rate.

Equations (7) and (9) can be combined to describe the magnitude and spatial distribution of the vertical velocity field as a function of the vertical distance above the orifice, x and the perpendicular distance y from the manifold. The general distribution depicted in Figure 1 applies to the line bubbler case as well, but with a change in definition for the horizontal axis.

Tuthill and Stockstill (2005) presented a rearranged form of equation (9)

$$U_c(x) = K \sqrt{\frac{-\ln(1 - x/h^*)}{(x + x_0)}} Q_a^{0.35} \quad (12)$$

where the term K is a constant approximately equal to $12.11 \text{ m}^{0.8} \text{ s}^{-0.65}$.

$$K = \sqrt{\frac{P_{atm}}{\sqrt{\pi} \rho_w C_b C_c}} \quad (13)$$

They suggest that when $x = h$, and the solution is for the centerline vertical velocity at the water surface, the term $L = \sqrt{-\ln(1 - h/h^*)/(h + x_0)}$ shows only a small variation for a range of h between 1 to 20 meters which is the typical depth of interest for ice control bubblers. Expanding on this observation suggests that

$$U_c = KLQ_a^{0.35} \quad (14)$$

and the term KL could be considered a constant. Therefore $U_c \propto Q_a^{0.35}$ assuming the centerline vertical velocity is not particularly sensitive to the depth. Further inspection of the L term shows some variation with depth in the range of interest, on the order of 15% with a peak value at a depth of just over 4 meters (Figure 2).

Tuthill and Stockstill (2005) measured vertical velocities in the plume above air screens submerged in shallow (0.5-meter) and medium (2.3-meter, 6.5-meter) depth settings. The measured velocities agreed reasonably well with calculated values in cases where it could safely be assumed that the momentum from the air jet leaving the orifice was not a significant factor. In other words, in cases where the momentum jet is small (low airflow) or in cases where the water depth is significant enough that the effects of the momentum jet are dissipated well below the surface. There is no general guideline for when this assumption is valid, but for most practical installations in navigation settings, the depth of submergence is likely to be equal to or greater than the 6.5-meter test case. At this depth, even with a relatively high airflow rate of 0.68 SCMM/m, the estimate from the analytical equation was close to the vertical velocity measured just below the surface.

Horizontal Surface Velocity

For line bubbler air systems designed to move or arrest ice on the surface, the velocity of the surface current following the transition from vertical to horizontal flow is the primary concern. Ashton (1974) provided an analytical solution describing the decay of the surface velocity with distance from the manifold's axis.

Tuthill and Stockstill (2005) observed that surface velocity measurements made at existing air screens and experimental systems in a lab setting did not agree well with those predicted using the analytical equation. They proposed both a linear and power-law fit between unit airflow and the observed average surface velocities that performed reasonably well over a range of airflows.

$$V = 55.289Q_a + 7.495 \quad (15)$$

$$V = 49.041Q_a^{0.52} \quad (16)$$

where V is the average near-surface water velocity (cm/s) and Q_a is the unit airflow (SCMM/m). The average near-surface velocity is taken over the length between the first measurement adjacent to the plume and $2H$, twice the depth of manifold submergence.

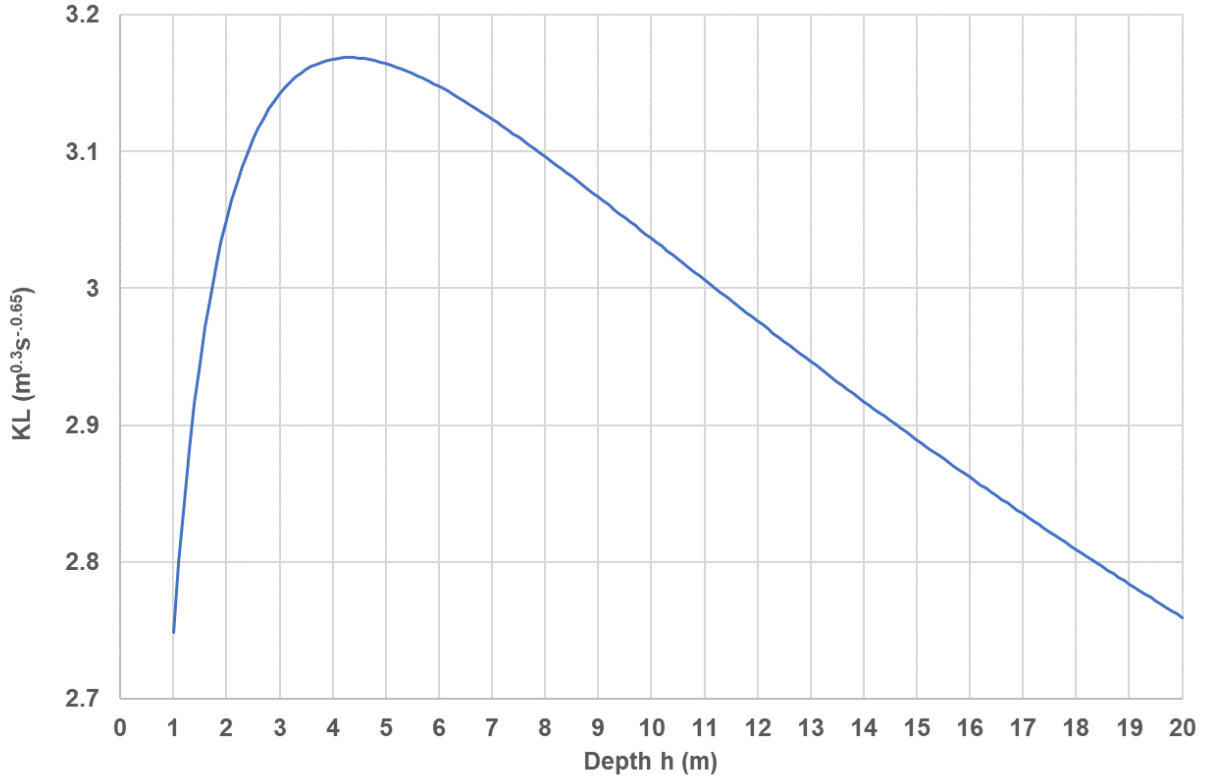


Figure 2: Relationship between KL term in Equation 14 and water depth.

Notably, the depth of water is not included in these relationships. The horizontal surface velocity is a function of the centerline vertical velocity in the plume. From inspection of the analytical solution (Equation 12), the vertical velocity is not strongly dependent on water depth above the manifold. This topic is addressed in the preceding section on vertical velocities.

Tuthill and Stockstill (2005) found that Equations (15) and (16) under predicted surface velocities in deeper water which was attributed to extended distance for flow development to occur. This makes intuitive sense because the buoyant air bubbles continuously provide a momentum flux to water column. However, the analytical solution assumes that this momentum is constantly transferred outwards to the linearly spreading plume. Equation (9) predicts a decline in vertical velocity at the surface with increasing depth beyond about 4.5 meters (at sea level), which is represented by the shape of the term KL vs depth in Figure 2.

Since the surface velocity does tend to increase as the depth of the bubbler increases, the actual flow conditions must vary somewhat from the idealized conditions assumed in the analytical case. The obvious place to look is just below the water surface where flow is characterized by divergent, horizontally flowing layers. Computational fluid dynamics (CFD) modeling of the Starved Rock air screen by Engel and Daly (2020) showed a turbulent oscillating flow pattern in the flow as it rose through the water column (Figure 3). The time-averaged vertical velocities matched the general linear spreading shape and gaussian velocity distribution of the analytical solution through most of the water column. However, the simulations showed turbulent

oscillations of a narrow, high-velocity stream that swung from upstream to downstream as it rose. As the airstream approaches the surface in the top 1.0 meters of flow, it was redirected alternately upstream and downstream in a shallow, fast-moving layer just below the surface. Consistent with this, Tuthill and Stockstill (2005) reported highly variable velocity readings in the field and therefore used time-averaged measurements and floating drogues to capture average flow conditions. It seems likely that while the analytical equations capture the time-averaged velocities of the plume as it rises, the plume behavior at the transition between vertical and horizontal flow is more complex than current time-averaged generalizations can capture. CFD modeling appears to be a promising path forward for analyzing these systems, however model setup and execution is time consuming. Currently, CFD is typically used for cases where no empirical method is available to estimate velocities, such as for air screens in a cross-flow environment.

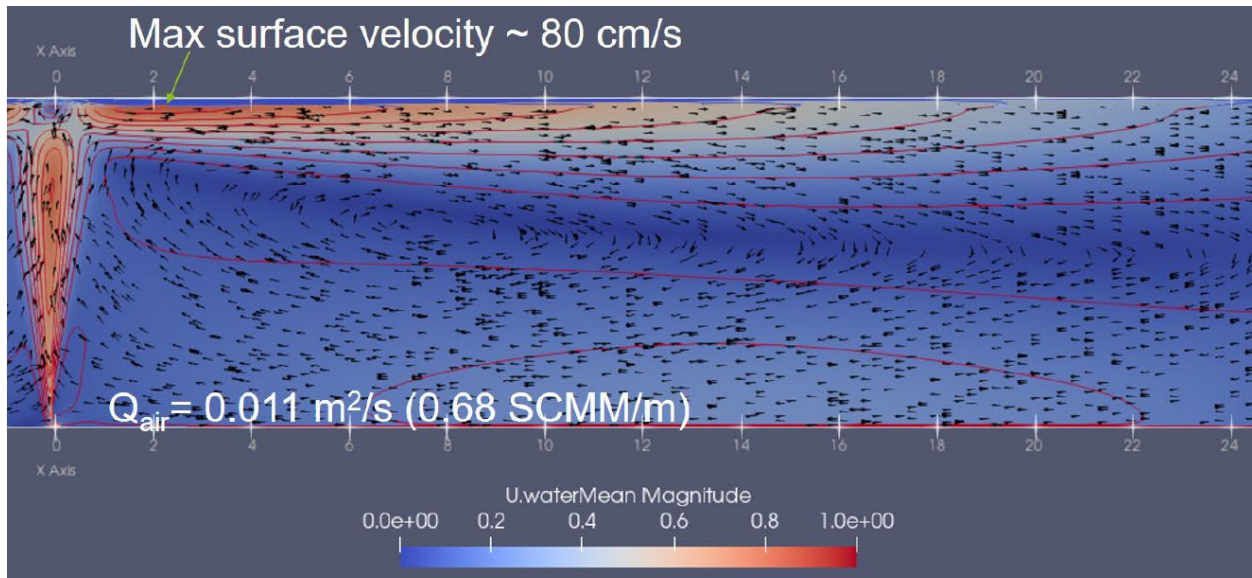


Figure 3: Water velocity magnitude plot from CFD model of air screen at Starved Rock Lock, (from Engel and Daly, 2020).

Recognizing the weak but apparent influence of water depth on surface velocity, Haehnel (2016) presented a modified form of Tuthill and Stockstill (2005) power-law relationship using additional experimental data from Hanamoto (1981):

$$V = \frac{(351 + 25.47H)}{10} Q_a^{0.4223} \quad (17)$$

In this form of the equation H is in meters, Q_a is in SCMM/M and V is in cm/s for consistency with the equations in Tuttle and Stockstill (2005).

Choked Flow

The design of an air system must include consideration of the effects of varying water levels, particularly for those installed within a lock. However, in many typical operational scenarios the

air discharge through the orifices is in a “choked flow” condition where discharge is not likely affected by downstream pressure variations.

When the pressure on the upstream side of an orifice is close to the downstream water pressure, the airstream is subsonic, flow through the orifice is subcritical and influenced by the magnitude of the downstream pressure. When the ratio of downstream to upstream pressure across an orifice falls below a critical threshold (for example by either by increasing the upstream air pressure, or lowering the downstream water level), the flow through the orifice reaches the local speed of sound and downstream pressure conditions do not influence the mass flow rate across the orifice. When the airstream becomes supersonic, the flow is supercritical and the mass flow rate is “choked” meaning it is controlled only by the upstream pressure and the geometry of the orifice.

Jobson (1955) provides mass flow rate equations for both subcritical and supercritical conditions for cases where the flow across the orifice is in an air-air system. Equation (18) is the general equation for mass flow rate.

$$\dot{m} = CK_N A \sqrt{P_0 \rho_0} \quad (18)$$

where C = the orifice contraction coefficient; K_N = a mass flow rate coefficient which varies with the flow condition; A = the projected area of the orifice; P_0 = the absolute upstream pressure and ρ_0 = the density of the air on the upstream side of the orifice. For subsonic flows:

$$K_N = \sqrt{\frac{2\gamma}{\gamma-1} r^{2/\gamma} \left(1 - r^{\frac{\gamma-1}{\gamma}}\right)} \quad (19)$$

where γ = the ratio of specific heats c_p/c_v (constant pressure and constant volume), which is 1.4 for air and r is the ratio of downstream to upstream pressure (P/P_0). The critical pressure ratio:

$$r_c = \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma}{\gamma-1}} \quad (20)$$

is a function only of γ , so for air $r_c = 0.5283$. Therefore, when the downstream pressure is less than 52.8% of the upstream pressure, the flow through the orifice is supercritical and Equation (19) does not apply. For supersonic flows the mass airflow rate coefficient is constant takes the form:

$$K_N = \sqrt{\gamma \left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}} \quad (21)$$

which is only a function of γ and therefore has a constant value. The general equation for mass flow rate (18) may be used with the mass flow rate coefficient calculated with either Equation

(19) or (21) depending on the pressure ratio r to yield a mass flow rate across an orifice in an air-air system.

The airstream's behavior in an air-water orifice system is potentially complicated by the presence of a much denser fluid than air on the downstream side. Leibson et al. (1956) conducted a series of experiments using a range of orifice diameters and orifice types to contrast the behavior of the air-water system with a classic air-air system. In the experiments, all orifices exhibited choked flow conditions, however the onset of supercritical flow occurred at a pressure ratio less than the critical ratio. When directly compared to previous experimental data, the sharp-edged orifices discharged less air than similar orifices in an air-air system. This was attributed to the influence of the formation of bubbles on the downstream side of the orifice and possible suppression of the *vena contracta* which occurs downstream of the orifice in an air-air system. For the experiments with round edged and thick plate orifices, the airstream behavior was similar to that observed in air-air systems. This was attributed to the *vena contracta* being contained within the orifice throat and thus not influenced by the water or bubble formation downstream. These results suggest that provided the orifice is thick enough to contain the *vena contracta*, consideration of the effects of water are not necessary and air-air equations can be applied.

The calculation of airflow rates is sensitive to the selection of C , the orifice contraction coefficient. Jobson (1955) provides equations for the theoretical quantity for the air-air system which varies with the pressure ratio, r and is a function of the incompressible contraction coefficient. The derivations of the theoretical equations assume that there is air on the downstream side of the orifice, which limits their direct application to the current problem where air is injected into water. In the case where the orifice is an ideal nozzle, $C=1$. The experimental results from Liebson et al. (1956) suggest variation in C with the pressure ratio however, that dependency is also a function of the orifice type. In all the air-water experimental results presented, C was around 0.9 as the flow approached choked conditions.

Liebson et al. (1956) presented their results in terms of C_n which is the ratio of the actual mass flow rate $\dot{m}$ to the theoretical rate under supercritical flow conditions. Equation (22) shows a slight rearrangement of the original form using a combination of Equations (18) and (21). It is important to note that for supercritical conditions C_n is effectively the contraction coefficient, but for subcritical conditions C_n also captures the ratio of theoretical subcritical flow to supercritical flow.

$$C_n = \frac{\dot{m}}{A \sqrt{P_0 \rho_0 \gamma \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma+1}{\gamma-1}}}} \quad (22)$$

Figure 4 shows the values of C_n versus the pressure ratio from the Liebson et al. (1956) experiments. The bold black line represents the range of C_n values that would be obtained using the theoretical air-air mass flow equations (Equation (18), using Equation (19) for K_N for subcritical conditions and Equation (21) for supercritical conditions). In the case plotted, C in Equation (18) is 0.95, which produced a curve that envelopes the experimental data. This shows

that C likely varies with the pressure ratio, orifice diameter and type. Precise calculation of a function for C for a specific orifice or nozzle would require experimental testing under the conditions it would be exposed to in operation.

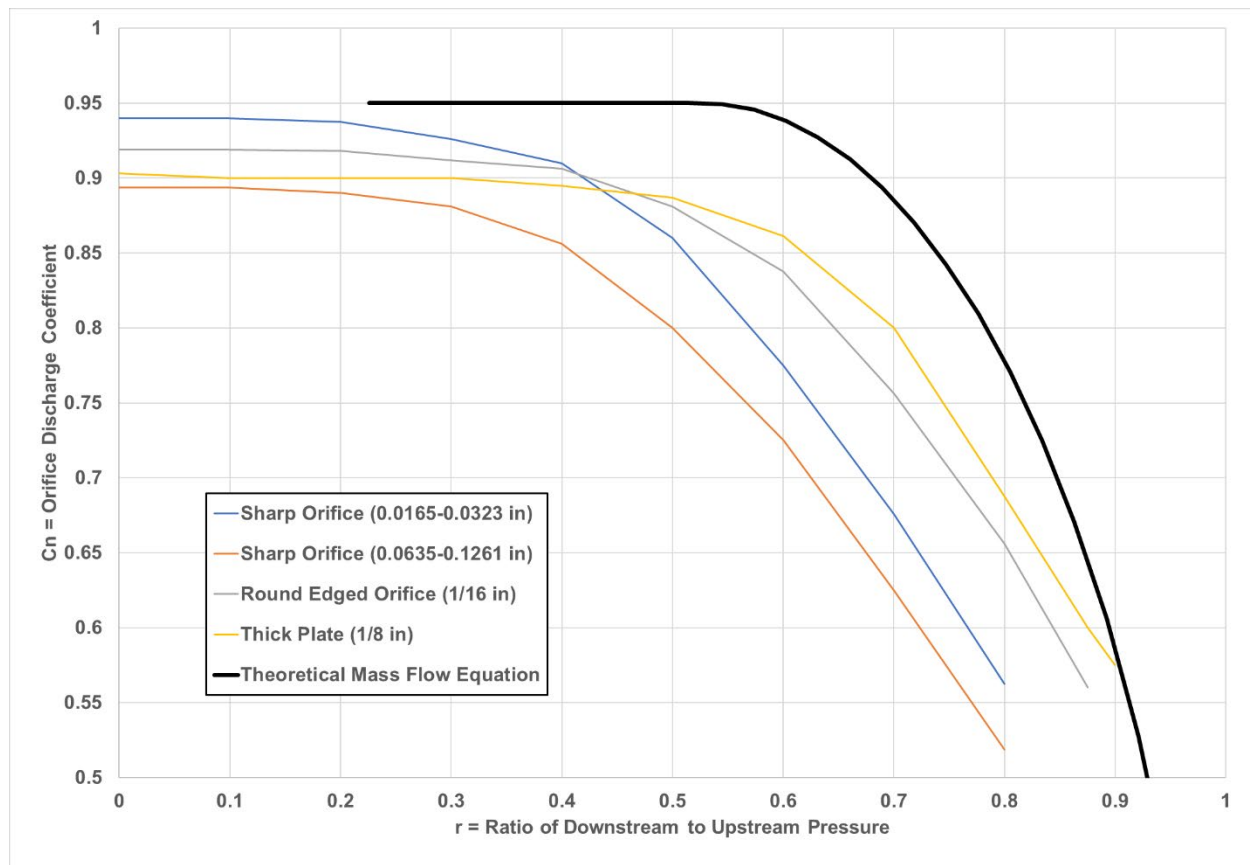


Figure 4: Comparison between C_n values experimentally measured by Liebson et al. (1956) and theoretical results from equation (18) using $C=0.95$.

The CRREL software BUB300 was updated as part of this project to allow the user to check for choked flow and use either subcritical or supercritical flow equations to calculate orifice discharge rates. The user can specify an orifice coefficient or use the default, $C=1.0$.

High Flow Air Systems

Point Bubblers

Point bubblers are the most basic form of air system used for ice control. They can be as simple as an open-ended pipe or fixed nozzle attached to supply line. A manifold with orifices that are spaced well apart will behave as a series of point bubblers.

High flow point bubblers are typically used to flush ice from gate recesses and disburse ice from a point location. Often point bubblers are simply open-ended 0.5 to 1.5 inch diameter pipes, a design that limits operational flexibility. In this configuration bubbler must be run at a relatively

low pressure to prevent excessive air consumption. The use of a nozzle or orifice on the end of the pipe will allow a range of operating pressures and corresponding airflow rates

Air Screens

Air screens or air deflectors are commonly used to prevent the movement of brash ice driven by vessels moving in a navigation channel. The system consists of an air manifold or diffuser pipe installed on or near the bottom of the channel. The manifold may be perpendicular to the channel and vessel traffic or in some cases it is placed at an angle to deflect ice in a desired direction.

Surface Velocity

The horizontal surface velocity produced by the air screen creates the drag force that holds or deflects the ice. The successful design of an air screen for this purpose requires generating a large enough current to oppose the forces driving the ice into the screen. The expected surface velocity generated by the screen can be calculated using empirical Equation (17), or in more complex cases using a CFD or physical model. The ice is held back if the forces acting in the upstream direction equal or exceed the forces acting on ice in the downstream direction:

$$f_{w(ds)} + f_a + f_g + f_{d(ds)} + f_i + f_v = f_{w(us)} + f_{d(us)} \quad (23)$$

where $f_{w(ds)}$ = water shear acting on the underside of the ice accumulation in the downstream direction

f_a = wind drag on the ice cover

f_g = downslope component of the gravity force on the ice accumulation

$f_{d(ds)}$ = water drag on the upstream face of the ice cover

f_i = force from ice impacting the upstream face of the ice cover

f_v = force from vessel passage

$f_{w(us)}$ = water shear acting on the underside of the ice accumulation in the upstream direction

$f_{d(us)}$ = water drag on the upstream face of the ice cover

The dominant forces in this balance are $f_{w(ds)}$, f_a , and f_v ; the shear from flowing water under the ice, wind acting on the surface and effects of vessel traffic. Wind and vessel traffic forces are transient, while flow generated by the air screen and ambient flow are usually relatively constant. The effects of vessel traffic on the air screen's ability to hold back ice can be very complicated. Vessels passing through accumulated ice will disrupt a steady-state balance, likely driving some ice through the air screen. However, the specifics of this interaction are functions of the vessel shape, speed and direction.

In a low-current environment typical to a lock approach, the dominant steady-state force is due to the shear generated by the air screen induced flow acting on the underside of the accumulated ice:

$$f_w = \frac{1}{2} C_d \rho_w A V^2 \quad (24)$$

where C_d = the under ice drag coefficient; ρ_w = the density of water; A = the under ice area the flow is acting on; V = the under ice water velocity. The force is proportional to the square of the velocity, a non-linear relationship meaning that the air screen gains increasing strength per unit increase in current velocity. Recalling that horizontal surface velocity is approximately proportional to the square root of the airflow rate (equations (16) and (17)), it can be deduced that the drag force generated by an air screen is linearly proportional to the airflow rate:

$$f_{w(us)} \propto Q_a \quad (25)$$

This means that, for example, doubling the airflow rate discharged by an air screen will roughly double its holdback capacity. This is a simplified rule of thumb that does not consider the effects of the channel walls or the effects of strong crossflow current.

Orifice Spacing

Existing guidance for selecting orifice spacing in design is limited and consists of various suggestions such as 2.4 meters (USACE 2018) or $\frac{1}{2}$ the water depth (Ashton 1974) without specific rationales. Spacing the orifices too far apart will result in the air screen performing as a series of point bubblers which may have dead zones of lower surface velocity between them that could allow ice to pass through the barrier. Provided the orifices are spaced close enough, the performance of a high-flow air screen, measured by the magnitude of the horizontal velocity on the surface, is a primarily a function of the average airflow rate per-unit-length of the manifold. Spacing the orifices too close together is not detrimental to developing a uniform surface velocity field but may unnecessarily increase construction costs with negligible performance benefit.

Since the vertical velocity field in the plume spreads linearly (Figure 1), the amount of interaction between plumes can be assessed graphically by plotting the standard deviations of adjacent plumes. Figure 5 shows the standard deviation of vertical flow in a series of plumes resulting from orifices spaced 8 feet (2.44 m) apart. These are point bubbler spreads overlaid with one another, so this method does not account for the effects of interaction between plumes. The interaction of adjacent plumes is complex and empirical methods to assess the effects of overlap are not available. In the case of Figure 5, the screen will likely have weak zones in the gaps between the central portions of the plumes. Reducing the spacing will increase plume interaction, approaching the limit case from Kobus (1968) where the orifices form a continuous slot in the manifold.

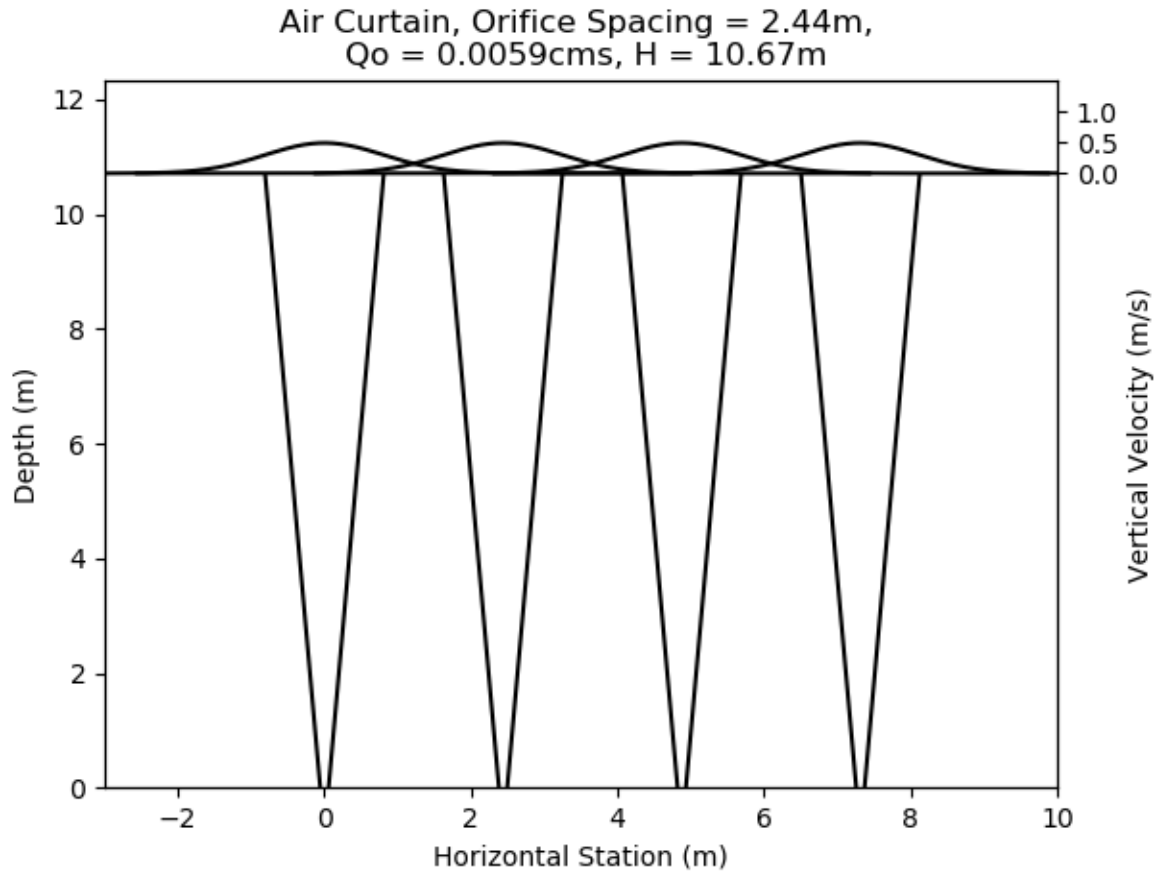


Figure 5: Example of 2D plume spread along a line bubbler, demonstrating potential for weak zones between plumes due to orifice spacing.

A potential criterion to define the maximum spacing of orifices is the distance at which the standard deviations of adjacent point bubbler plumes would no longer overlap. This is a somewhat arbitrary selection, but is relatively easy to define and likely to result in a conservative design. The spacing would be twice the standard deviation distance b , defined in equation (11) at the water surface:

$$s = 2(H + x_0) C_c Q_a^{0.15} \quad (26)$$

where H is the water depth, C_c = a coefficient measured experimentally ($0.152 \text{ m}^{-0.45} \text{ s}^{0.15}$), Q_o is the orifice airflow rate in cubic meters per second at standard temperature and pressure (SCMS), and x_0 = distance from the analytical origin. x_0 is a correction term accounting for the initial momentum of the air jet and the complex flow behavior before the region of induced flow.

Figure 6 shows the regions of bubbler behavior as a function of spacing vs the amount of airflow. The spacing is normalized by dividing it by the $(H + x_0)$ term. It can be seen that by using this criteria, the maximum orifice spacing is approximately equal to 10-20% of the water depth with closer spacing recommended for lower flow conditions.

Uniform horizontal surface velocity is the primary objective when selecting an appropriate orifice spacing and this method doesn't directly address this. Future CFD modeling or physical modeling could refine the answer to this question, however using the maximum spacing with equation (26) will likely provide reasonable and conservative results.

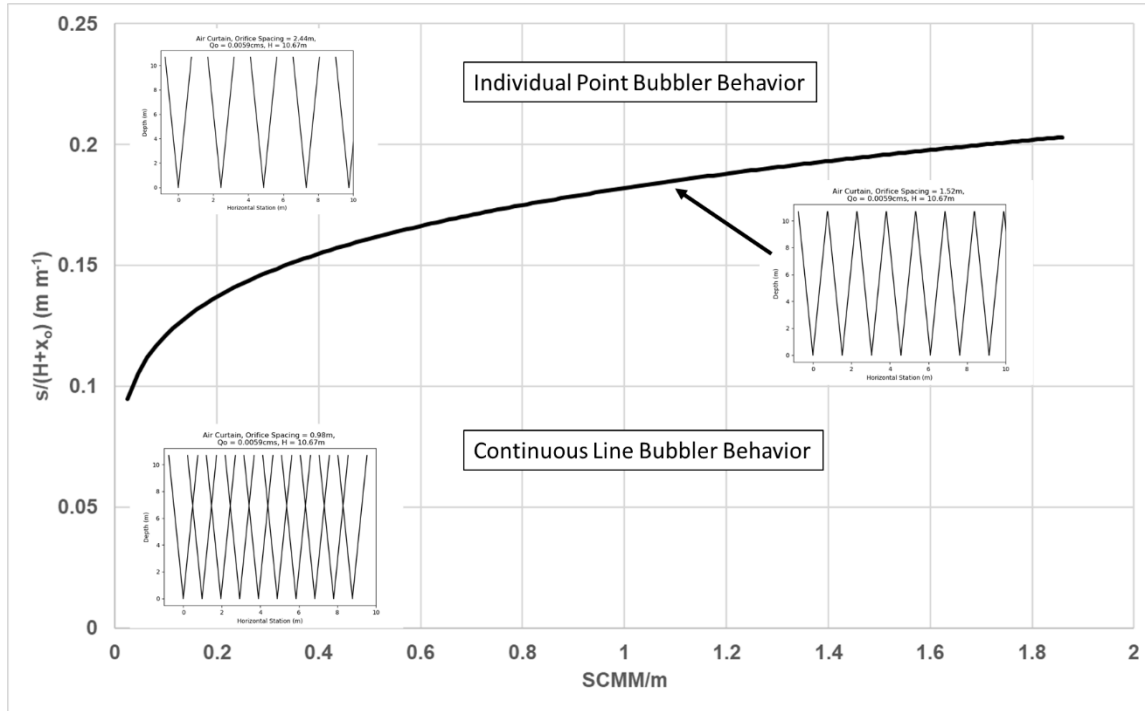


Figure 6: Zones of line bubbler behavior as a function of airflow rate (x-axis) and orifice spacing, s , normalized by water depth, H and the distance to the analytical origin, x_0 (y-axis). Systems plotting above the bold line will tend to behave as a series of isolated point bubblers while those plotting below will tend to produce a highly interactive air screens.

Orifice Diameter

For a given pressure in the manifold and downstream water level, the airflow rate is linearly related to the area of the orifice (equation (18)). Therefore, airflow rates are directly proportional to the square of the orifice diameter.

Larger diameter orifices ($\frac{1}{4}$ to $\frac{1}{2}$ inch) are less prone to being plugged with debris, but will discharge higher volumes of air. Depending on the air supply capacity of the compressor or air storage, manifolds with large orifices can be run at very low pressures to prevent the air screen demand from exceeding the available air supply. This limits the range of operation of an air system. Orifices with smaller diameters ($\frac{1}{8}$ – $\frac{1}{4}$ inch) can still provide high airflow rates at higher pressures, but also allow airflow to be throttled to lower flow rates as desired. When the local water is highly turbid or biological growth is significant, caution should be used when using small orifices.

Pipe Dimensions

Minimizing excessive pressure drop in pipes is the primary design consideration in a conceptual design of an air screen due to the high airflow conditions. This applies to both supply pipes and the manifolds discharging air. The pressure drop in a pipe segment is approximated using the Darcy-Weisbach Equation:

$$\Delta P = f \frac{L}{D} \frac{\rho v^2}{2} \quad (27)$$

where L = the length of the pipe segment; D = the diameter of the pipe; ρ = the density of the air; v = the velocity of the air and f = the Darcy friction factor. For turbulent flows, when the dimensionless Reynolds number, Re exceeds 4,000, a certainty for air high flow sir systems, f is a function of the Re , the pipe diameter and the wall roughness. Following Moody (1944) the empirical Colebrook function is commonly used to solve for f :

$$\frac{1}{\sqrt{f}} = -2 \log_{10} \left(\frac{\frac{\epsilon}{D}}{3.7} + \frac{2.51}{Re\sqrt{f}} \right) \quad (28)$$

where ϵ = the relative pipe roughness. This is a transcendental equation that requires empirical approximation to solve for f .

Supply Pipes

There is not a specific criterion for allowable pressure drop in a supply line, but clearly the pressure delivered to the manifold must be sufficient to allow the screen to operate at the range of design flow rates. USACE (2018) suggests that supply lines be 3 or 4 inches in diameter, where larger diameters are more appropriate for longer supply lines. It is not difficult to calculate the pressure drop in a supply line and the airflow from the compressor to the manifold should be included in the system analysis.

It is recommended that supply pipes include a check valve below the normal water surface to prevent water from backing up into the supply line during periods that the system is not operating. In frigid conditions standing water in the line near the water surface can freeze damaging the supply pipe (USACE, 2018).

Manifold Pipes

For the manifold design, the orifice sizes must be compared with the manifold diameter to ensure that the flow rates out of the orifices are as nearly uniform as possible along the length of the manifold. The flow rates out of the orifices will be nearly uniform if the sum of the area of all the orifices is less than 25% of the cross-sectional area of the manifold

$$\sum_{j=1}^n \pi \frac{d_j^2}{4} < \frac{1}{4} \pi \frac{d_{manifold}^2}{4} \quad (29)$$

where d_j = the diameter of orifice j ; and $d_{manifold}$ = the diameter of the manifold (USACE, 2018). When all of the orifices are the same diameter this reduces to:

$$2\sqrt{n} \cdot d_{orifice} \leq d_{manifold} \quad (30)$$

A complete analysis should include the calculation of pressure and discharge along the entire length of the manifold. If the pipe diameter is too small or if there are too many orifices, a large pressure differential may exist across the manifold resulting in non-uniform flow, with a potentially weak far end. The coefficient of uniformity (Haenhel, 2016):

$$CU = (Q_{max} - Q_{min})/Q_{ave} \quad (31)$$

provides a measure of the variation of the flow across the manifold, assuming all the orifices are the same diameter. The closer CU is to zero, the more uniform the flow across the air screen. Since the under-ice drag force generated by an air screen surface current is linearly related to the airflow rate (Equation (25)), the coefficient of uniformity can also be used as a measure of the variation in ice holdback force along the length of the manifold. If airflow is strong on the end nearest the compressor, and weak on the far end, ice may be sufficiently held back by the strong part but be able to pass through the weak zone. There are no specific guidelines for a maximum value of CU , but for a new design it should as close to zero as practical (~ 0.01) and for retrofits generally less than 0.15.

The typical practice used by USACE-CRREL to size the components of a new air system is to evaluate the performance of an air system geometry using an iterative algorithm. A method is described in the Ice Engineering Manual (USACE, 2018) however the particulars of that method differ from those used in the software program BUB300. The method used by the solver are summarized here. An upstream compressor pressure is specified and a trial total air flow rate is selected. The pressure drop in the supply line is calculated using the trial airflow and equation (27). The pressure drop is subtracted from the initial pressure and the remaining pressure is used to calculate the airflow at the first orifice. The airflow is calculated using equation (18) (in the updated version of the software checks for and accounts for choked flow conditions at the orifice). The air flow rate through the orifice is subtracted from the air flow rate upstream of the orifice and the remaining airflow is then used to calculate the pressure drop in the next segment. This process continues for each orifice and manifold segment, until the end of the line. The total sum of airflow leaving through the orifices is compared with the initial trial air flow rate. The difference between the trial and calculated total airflow rate is used to update the trial airflow rate (30% of the difference is subtracted) and the routine is run again. When the difference between the trial and calculated total airflow rates fall below a user specified threshold, the solution is considered to have converged and the calculated airflow rates for each orifice represent the expected discharge for that pressure.

Orifice Pinch Valves

Some air system installations on USACE projects have included rubber pinch valves (also called duckbills) on the orifices (USACE, 2013). The pinch valves prevent intrusion of water into the manifold when the system is not operating, preventing sediment deposition in environments with turbid water. The effect of the pinch valves on system performance has not been studied, but for high-flow air systems they likely cause only a minor reduction in airflow at each orifice.

Drain holes

Manifolds should be equipped with at least one drain hole near the far end of the system, and potentially more at low points in the system depending on the manifold layout. Drain holes will allow water to be more rapidly purged from the system on start-up and will allow the manifold to fully drain during operation. The drain hole diameter should be on the order of the typical orifice diameter on the manifold.

Air Screen Performance

The primary purpose of an air screen is to create a horizontal current to stop brash ice from being transported downstream. The performance of an air screen is typically quantified using the induced horizontal velocity which can be approximated using an empirical relationship between the supplied airflow, manifold length, and operating depth (Equation 17). The actual hold back performance is related to the under-ice shear force exerted by the air screen's induced horizontal velocity (Equation 24) compared to the total driving force acting on the ice (current, wind, vessel traffic) per unit length of the span of the channel.

The type of air screen discussed in this report is made up of a line of individual bubble plumes that usually interact and form a relatively uniform barrier. Orifice spacing and operating depth are the key parameters that influence the interaction between adjacent air plumes (Equation 26). Large spacing will result in less plume interaction and might result in weak zones or holes in the air screen. Orifice spacing between 10% and 20% of the water depth should provide sufficient overlap between adjacent air plumes.

The performance of an air screen relative to another design can be achieved by comparing the relative hold back velocity by rewriting Equation 24.

$$\frac{f_{w,new}}{f_{w,ref}} = \left(\frac{v_{new}}{v_{ref}} \right)^2 \quad (32)$$

Where v_{new} is the hold back velocity of the new design and v_{ref} is the hold back velocity of the reference design.

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